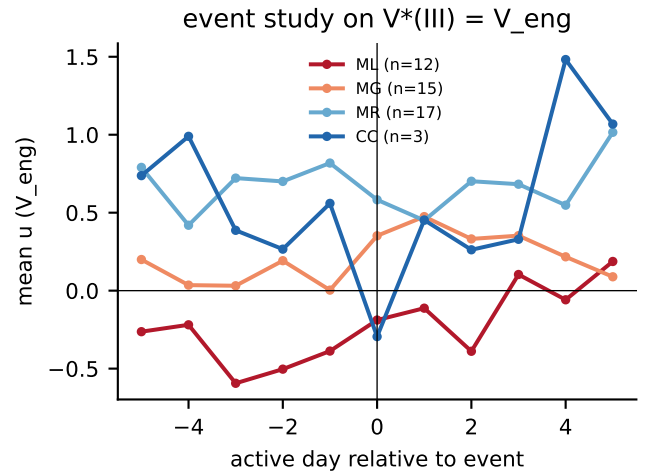
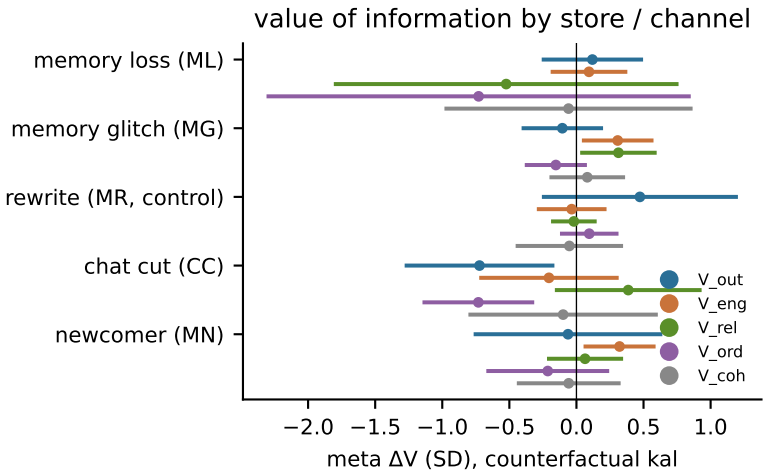
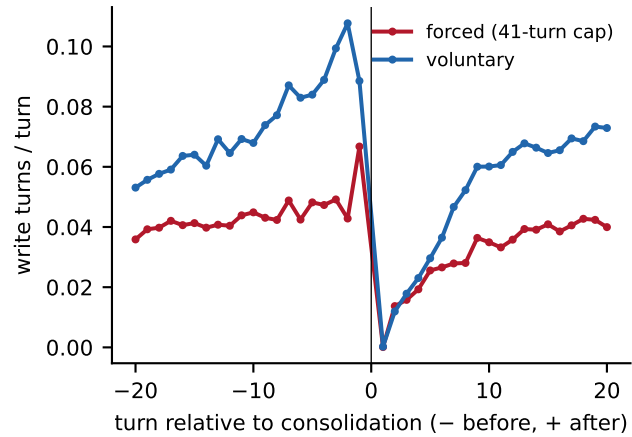


H15 • Semantic information through natural scrambles (round 1, non-holdout)

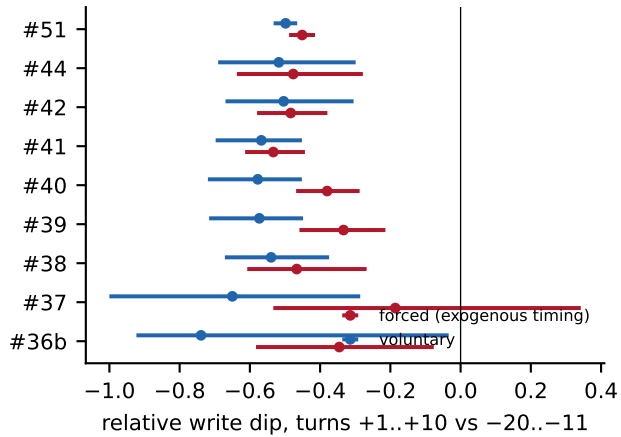
Question: which information keeps LLM agents viable? Kolchinsky-Wolpert: information is semantic if scrambling it lowers viability V. Natural scrambles (non-holdout): 16 memory losses beyond normal consolidation (ML), 84 rewrites (MR, control), 20 newcomers with empty memory (MN), 10 chat cuts (CC), 18.6k forced and 12.2k voluntary context erasures at consolidation (CF/CV). Unit = goal period. V chosen by D2.6 (restored after goal-change shocks): INCONCLUSIVE in both regimes -> V*(I) = reliability, V*(III) = engagement. Synthetic (real village skeleton): null calibrated (5%); power 41% for a -0.4 SD memory-loss effect; selection biases kal -0.10; a transient slump around a reset biases it -0.57 and the pre-trend catches only ~30% of those. Day-scale memory: ML on V* -0.55 [-1.78, +0.68] (2 periods; engagement alone +0.09 [-0.19, +0.37]); newcomers +0.15 [-0.18, +0.47], engagement +0.32 (z 2.3, novelty). Rewrites ~0 (control passes). => no detectable day-scale cost of losing stored memory (P1, P4 fail). Context window: P5 as registered FAILED (forced - voluntary dip = +0.010 writes/turn, z 5; voluntary consolidations follow write bursts). Post-hoc exogenous-timing estimate: writes in the 10 turns after a forced erasure fall 33-53% vs turns -20..-11 (8/9 periods, meta -0.44). The amount written to memory does not reduce the dip (rho ~ 0): the load-bearing information is in the context and is re-acquired from the environment within ~15 turns, not carried by consolidation. Chat cuts: 3 usable events, output -0.72 (z -2.4), selection-prone. Reading: eta(memory, day scale) ~ 0 within power; the semantic information that keeps output going lives in the session context.



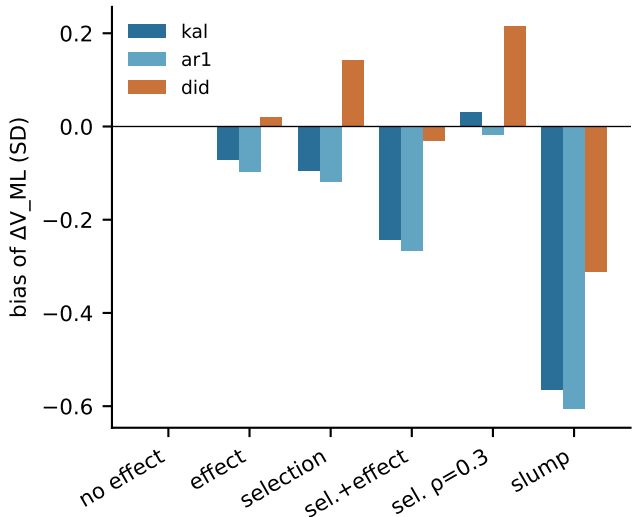
context erasure: write rate around consolidations (regime III)



POST-HOC: cost of a context erasure, per period



ML estimator bias (150 reps, real skeleton)



power at village sample sizes

